What changes an asteroid’s path?

antocciphysics

UNIT 2 / DAY 1 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain how gravity can change velocity without contact.

SUCCESS CRITERIA - I can...

1. Draw velocity tangent to a path and force toward the attracting body.
2. Distinguish motion from a change in motion.
3. State what a straight-line model leaves out.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A simplified model predicts position (-30, 50) km with bounds ±3 km in x and ±5 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.

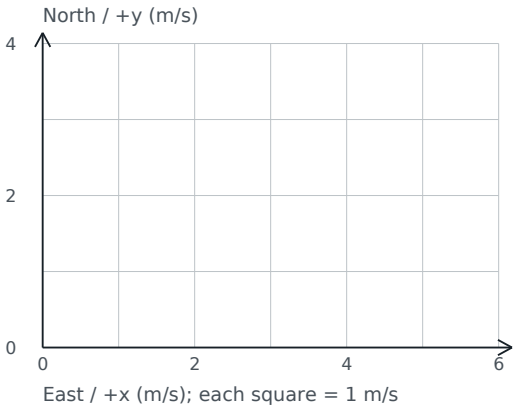
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a = F_{net}/m$.
Gravity attracts masses and can change speed, direction or both. Our asteroid scenario is fictional.

Worked reasoning: A fictional asteroid has a velocity arrow pointing right. A brief upward force gives it an upward change in velocity. The new velocity combines the original rightward motion with this change.

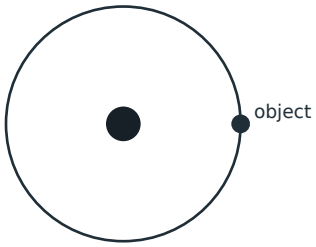
Your turn: A spacecraft is initially moving right. A short thruster burn adds an upward change in velocity. Draw the original velocity and the change head-to-tail. Draw the final velocity from the starting tail to the ending tip.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Sketch the velocity and gravitational-force arrows at the marked point on the orbit. Why do they point in different directions?

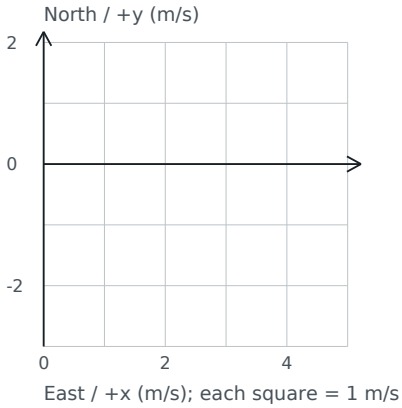


The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

A simulated probe starts with velocity (4, 0) m/s. Its thruster produces acceleration (0, -1) m/s² for 2 s. Plot the initial and final velocity vectors. Find the final speed and explain whether speed stayed constant.



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

05 Explain your reasoning

A student says space has no gravity because astronauts float. Explain how orbiting astronauts can be falling.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

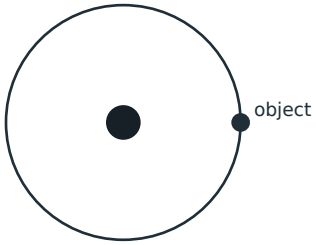
A 0.5 kg system accelerates at 0.7 m/s². Predict acceleration for 0.9 kg at the same net force. Explain the effect of greater resistance.
Use the table to keep the controlled and changed quantities distinct.

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

07 Exit check

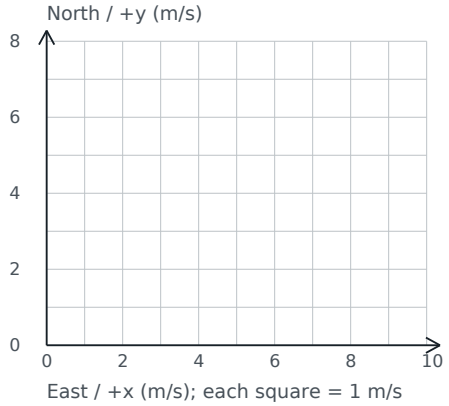
A force stays perpendicular to velocity during ideal circular motion. Explain what changes and what stays constant.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A simulated probe starts with velocity (8, 0) m/s. Its thruster produces acceleration (0, 2) m/s² for 3 s. Draw the change in velocity head-to-tail with the initial velocity. Find the final components and speed.



Marzano check - circle: 1 2 3